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Instruction Granularity as Planning Width: Formalizing and Benchmarking for Vision-Language-Action Models

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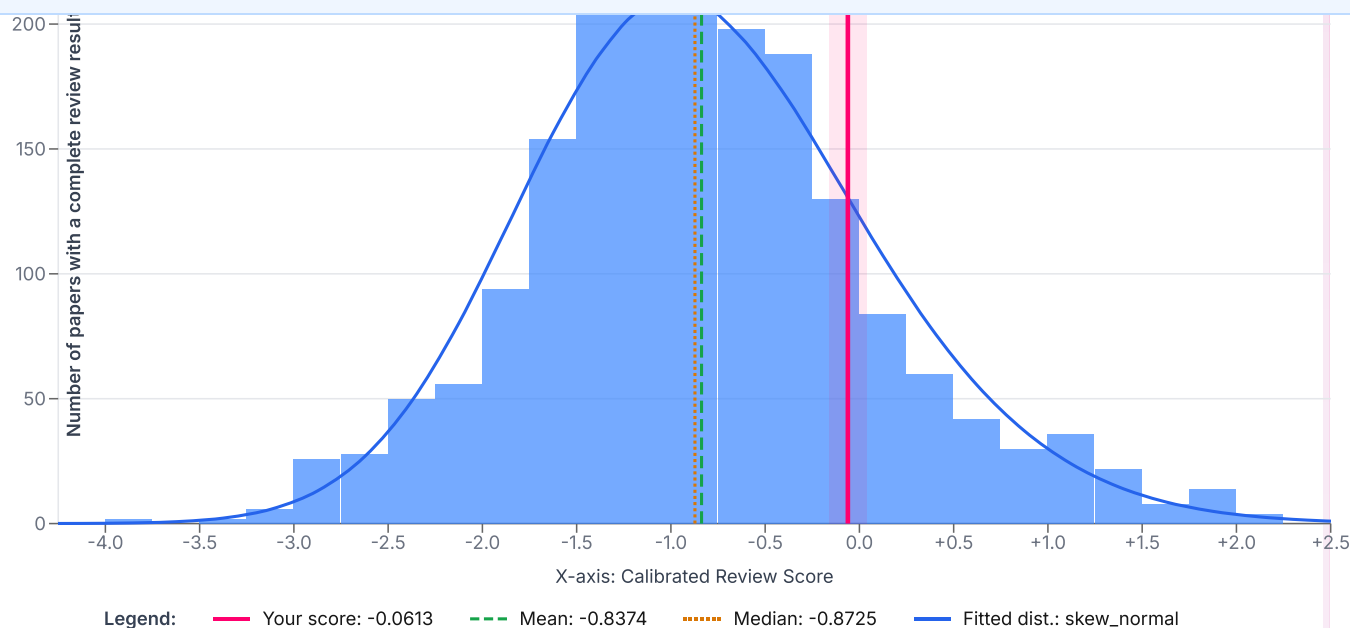
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Desk Rejection Assessment:

Paper Length

Pass ☒. The paper is within the standard length for a main track submission.

Topic Compatibility

Pass ☒. The paper's focus on Vision-Language-Action (VLA) models, instruction grounding, and embodied AI aligns perfectly with the ACL topic "Multimodality and Language Grounding to Vision, Robotics and Beyond."

Minimum Quality

Pass ☒. The paper is well-structured, written in clear English, and presents a coherent research narrative with a methodology, a new benchmark, experiments, and analysis. The claims appear to be scientifically grounded, and the paper meets the minimum quality standards for a top-tier conference review.

Prompt Injection and Hidden Manipulation Detection

Pass ☒. A review of the paper content reveals no hidden prompts, invisible text, or other attempts to manipulate the review process.

Expected Review Outcome:

Paper Summary

This paper introduces a formal, agent-centric definition of instruction granularity for language-guided embodied AI. The authors propose to quantify granularity using "rule width" (\$w\$), a metric from width-based planning that measures the number of state features an agent must concurrently track. To study this variable in a controlled setting, they introduce the Mini-BEHAVIOR-Gran benchmark. Through experiments on this benchmark, they uncover a non-monotonic, U-shaped relationship between granularity and model performance, where both very fine-grained and very coarse instructions outperform those in the middle. They

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This is a strong and thought-provoking paper that makes several valuable contributions to the study of language-guided embodied agents.

First, the primary strength is the formalization of instruction granularity as rule width ($\$w\$$). The field has long struggled with fuzzy, text-centric definitions of granularity (e.g., sentence length). By grounding the concept in an agent-centric, task-level property borrowed from classical planning, the authors provide a principled and quantifiable metric for the decision-making burden an instruction imposes. The way width is computed and explained using the running example in Section 3.1 and Table 1 is clear and intuitive. This formalism has the potential to become a standard tool for analyzing instruction-following behavior.

Second, the introduction of the Mini-BEHAVIOR-Gran benchmark is a significant contribution. By systematically varying instruction width ($\$w\$$) while keeping the underlying tasks, visual environment, and action space constant, the authors have created a much-needed controlled testbed. This allows them, and future researchers, to isolate the effect of granularity from confounding factors like perceptual difficulty or control complexity, a limitation in many existing benchmarks. The experimental design shown in Figure 1 is sound and enables a rigorous investigation.

Third, the empirical findings are insightful and non-obvious. The discovery of the U-shaped performance curve (Figure 3) is particularly interesting. The intuitive assumption would be a monotonic decrease in performance as instructions get coarser (i.e., width increases). The paper's finding that performance rebounds at the coarse end challenges this simple view and reveals a more complex dynamic.

Finally, the paper provides a compelling, granularity-centric analysis of the "language-ignoring" phenomenon. By connecting the rebound in performance for coarse instructions to an increased likelihood of the model bypassing language (Section 6), the authors offer a new perspective on a critical failure mode in VLA models. The analysis of generalization asymmetry in Figure 4, which shows that coarse-to-fine transfer is more effective than the reverse, and the proposal of mixed-granularity training are practical and well-supported takeaways for the community.

Summary Of Weaknesses

Despite its strengths, the paper has several weaknesses that temper its conclusions and should be addressed to strengthen the work.

- 1. The "Proxy" Nature of Width for Neural Models:** The central premise is that symbolic rule width ($\$w\$$) is a "rigorous proxy for the internal grounding demand in learning-based VLA models" (Abstract, Page 1). While the experiments in Figure 6 show a clear *correlation* between width and success rate, the paper does not provide direct evidence for the *causal mechanism*. It is asserted, but not proven, that the VLA model's struggle is due to an inability to "track multiple simultaneous feature dependencies" (Section 5, Q3). The width metric is derived from a symbolic state abstraction ($\$\Phi\$$) that the neural model does not explicitly see. It is possible that width is simply a good correlate for another underlying factor, such as the linguistic complexity of the template-generated instructions or the ambiguity of the required action sequence, which is the true cause of the performance drop. The paper needs to be more cautious with its claims about what the VLA model is *internally* doing.
- 2. Artificiality of the Experimental Setup:** The controlled nature of the Mini-BEHAVIOR-Gran benchmark, while a strength for isolating variables, is also a significant limitation on the generalizability of the findings. The entire framework relies on (a) a manually defined, symbolic feature set $\$\Phi\$$ to compute width, and (b) template-based language generation. This begs the question of how this framework would apply to real-world scenarios with unconstrained natural language and raw sensory inputs, where a pre-defined symbolic state is not available. The paper's claim that width is a "cross-task yardstick" feels overstated when its computation is tied to this highly structured, artificial setup.
- 3. Inconclusive Evidence for the Language-Ignoring Mechanism:** The analysis in Section 6, which attempts to explain why coarse instructions lead to language ignoring, is intriguing but ultimately speculative. The paper invokes a theory from Xu et al. (2025) that high predictability $\$P(I|v)\$$ encourages language ignoring. However, it provides no direct measurement or estimation of $\$P(I|v)\$$ across granularity levels. The attention analysis in Figure 9 is presented as a probe but is inconclusive, as the authors admit there is "No clear trend across granularity levels." In fact, the attention data could even be interpreted as weakly

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the number of rule calls in Figure 5. However, it fails to control for other potential confounders introduced by the template-based language generation. For instance, the fine-grained instruction example given on page 5 ("If you are holding the rag but the rag is not yet soaked, soak the rag") is syntactically more complex (e.g., nested clauses) than a coarse instruction might be. The performance dip at medium granularity could be related to a "worst of both worlds" scenario in linguistic structure, not just planning width. The claim of having "isolated" width is therefore too strong.

5. **Limited Scope of the Benchmark:** The use of Mini-BEHAVIOR, a simplified 2D grid-world environment, limits the scope of the conclusions. The tasks, while long-horizon, do not involve the complex 3D perception, physics, and continuous control challenges present in more realistic simulators like iGibson, AI2-THOR, or Habitat. It's plausible that the observed U-shaped curve is an artifact of this simplified domain and may not hold in settings where perceptual and control errors are more dominant factors.
6. **Interpretation of Generalization Asymmetry:** In Section 5.2, the paper interprets the generalization asymmetry in Figure 4 (coarse-to-fine > fine-to-coarse) as evidence that coarse-trained models learn "robust language grounding". An alternative, and perhaps more cynical, interpretation exists: the coarse-trained models simply learn a better vision-based policy that is *less dependent* on language, consistent with the language-ignoring findings in Section 6. Therefore, when tested on fine-grained instructions, the model succeeds not because it's grounding them well, but because its policy is already good enough without relying on them. The claim of "robust grounding" needs to be disentangled from "robust vision-only policy".
7. **Incomplete Discussion of Related Work:** The paper misses several recent and highly relevant works on benchmarking planning and instruction-following for VLA models. This omission weakens the paper's positioning and its claim of providing the "first" controlled benchmark for this specific problem. Please see the next section for a detailed list.

Potentially Missing Related Work

The paper would be strengthened by discussing and positioning itself with respect to the following highly relevant works, which were not cited:

1. **Su, Y., Ling, Z., Shi, H., "ActPlan-1K: Benchmarking the Procedural Planning Ability of Visual Language Models in Household Activities", 2024.** — This paper introduces a benchmark specifically for evaluating procedural planning in household activities, a domain very similar to Mini-BEHAVIOR. It should be discussed in Section 4 when motivating the need for Mini-BEHAVIOR-Gran and comparing against existing benchmarks.
2. **Wang, D., Li, X., Zhang, Z., "ProcWorld: Benchmarking Large Model Planning in Reachability-Constrained Environments", 2025.** — ProcWorld is another recent benchmark focused on evaluating planning for large models. Its focus on reachability constraints is conceptually related to the state-feature dependencies captured by rule width. This should be discussed in Section 2 or 4.
3. **Zhang, S., Qiao, Y., Wang, Q., "FlexVLN: Flexible Adaptation for Diverse Vision-and-Language Navigation Tasks", 2025.** — This work on Vision-and-Language Navigation explicitly deals with adapting to diverse tasks and instructions, which is related to the theme of handling different granularities. It would be relevant for the discussion on generalization in Section 5.2.
4. **Fang, I., Zhang, J., Tong, S., "From Intention to Execution: Probing the Generalization Boundaries of Vision-Language-Action Models", 2025.** — This paper also investigates the generalization capabilities of VLA models and introduces a benchmark suite. It provides a relevant point of comparison for the generalization experiments and claims made in this work.

Comments Suggestions And Typos

Thank you for this interesting and well-written paper. The ideas are stimulating. Here are a few suggestions and questions that I believe could help improve the manuscript further. My score could increase if the authors can provide convincing responses, particularly to points 1 and 2.

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T_s)? This would move the argument from correlation to more direct evidence.

2. **Addressing the Language-Ignoring Explanation:** The explanation for language ignoring hinges on the unverified assumption about $P(I|v)$. Could you provide any empirical estimate of this, even a rough one? For example, one could train a simple model to predict the instruction I from the visual state v and measure its accuracy across granularity levels. If the accuracy is indeed higher for coarse instructions, it would substantially bolster your argument in Section 6. Without this, the connection remains tenuous.
3. **Clarification on Failure Modes:** In Figure 7, you categorize tasks by failure mode (regression vs. stagnation). However, the y-axis is "Avg Step Ratio", which is not clearly defined in the paper. Please clarify what this metric represents and how it is calculated. Also, the link between width and this failure mode seems correlational. Is there a deeper reason why high-width tasks lead to stagnation (getting stuck) while long-horizon, low-width tasks lead to regression (undoing progress)? Expanding on the brief analysis in Section 5.2 would be helpful.
4. **Minor Typos/Clarifications:**
 - Page 3, Table 1 Caption: "Cleaning k Shoes $\left(\{k = 2\} \right)$ w/o instructions". This is a bit confusing. I believe you mean this is the width computation for the *underlying task*, before applying instructional decomposition. Please clarify.
 - Page 3, Section 3.3, first paragraph: "...a set of 'if-then' rules $\mathcal{R} = \{r_1, \dots, r_m\}$ defined over the feature space Φ . This rule-based representation does not aim to capture...". There is a line break after Φ that seems unintentional.
 - Page 4, Width Analysis of r_{1b} : The logic here is quite dense and critical to understanding the width calculation. It might benefit from being slightly rephrased for clarity, perhaps with a small diagram showing how maintaining a precondition (H_r) reduces the active search space.

Confidence

4 Quite sure after careful checking: I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness

4 Strong: This study provides sufficient support for all of its claims. Some extra experiments could be nice, but not essential. The core empirical findings appear solid, though some of the interpretations could be strengthened as noted in the weaknesses.

Excitement

4.0 Exciting: I would mention this paper to others and/or make an effort to attend its presentation in a conference. The formalization of granularity and the U-shaped finding are genuinely interesting and will likely stimulate discussion.

Overall Assessment

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4 Conference: This paper could be accepted to an ACL conference. It demonstrates sufficient novelty, soundness, and impact. The paper introduces a novel and valuable conceptual tool (width-as-granularity), supports it with a well-designed new benchmark, and delivers non-obvious empirical findings that have clear implications for VLA research. While there are weaknesses regarding the interpretation of the results and the artificiality of the setup, the strengths—particularly the new formalism and the core empirical discoveries—outweigh them. The work is a solid contribution that advances our understanding of instruction following in embodied agents.

Best Paper Justification

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focus on instruction granularity only). However, the paper would benefit from a more explicit "Limitations" section. Key limitations to discuss would be: (1) The reliance on the Mini-BEHAVIOR simulator, which simplifies perception and control, and how the findings might change in more realistic 3D environments. (2) The dependency on a manually-defined symbolic feature set Φ , which limits the applicability of the width metric to real-world, unstructured instructions. (3) The use of template-based language, which avoids the complexities of parsing diverse, natural human language.

Regarding societal impact, the work is primarily foundational research and has no immediate, direct societal impacts. Potential negative impacts are remote but could be related to the dual-use nature of autonomous systems. A brief statement to this effect would be appropriate.

Ethical concerns

None. The research is conducted in a simulated environment and does not involve human subjects, private data, or other common sources of ethical concern. The authors also include a statement about their use of an LLM for language polishing, which is good practice.

Needs Ethics Review

No.

Reproducibility

4 Mostly reproducible: They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method. The authors provide a good level of detail on their methodology, experimental setup, and model architectures (with references to specific models). They plan to release the benchmark, which is excellent. Given the code for the benchmark and the detailed description, the main results should be reproducible.

Datasets

4 Useful: I would recommend the new datasets to other researchers or developers for their ongoing work. The proposed Mini-BEHAVIOR-Gran benchmark will be a valuable resource for the community. It provides a controlled environment to study instruction grounding, which is a critical and under-explored area. Its value is high because it enables targeted, causal experiments that are difficult to perform with existing, more complex benchmarks.

Software

1 No usable software released. The paper does not state that the model implementation code will be released, only the dataset/benchmark.

Disclaimer: Only the main paper is reviewed; anonymity and formatting are not checked. The desk-rejection assessment evaluates content completeness, correctness, novelty, clarity, and scientific quality in line with common ACL review guidelines.

Reviewed on March 07, 2026  Gemini-2.5-Pro

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